

Supplementary Material for RAOP: Runtime-Adaptive Operator Selection for Routability-Driven Placement

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Operator implementation settings

Table S1 records the secondary engineering constants and candidate-selection rules used by every reported experiment. The main manuscript retains the operator-defining differences; this table provides the corresponding implementation detail.

Table S1: Fixed implementation settings for the four response operators.

Setting	Applicable path	Fixed value or rule
Legacy inflation field	Legacy	$(C + 1)^2$, clamped to $[1, 2]$ and projected to cells by overlap
Momentum update	Scalar, XY, no-deflation	Update t uses C^t and the following refinement/GR produces C^{t+1} ; $r_k^0 = 1$, $\Delta r_k^0 = 0$, $\Delta r_k^t = 0.4\Delta r_k^{t-1} + 0.6u_k^t$, and upper bound 2.0; scalar/no-deflation set each original dimension by $\sqrt{r_k^t}$
Stimulus denominator guard	Scalar, XY, no-deflation	ϵ is the machine epsilon of the local-congestion tensor's floating-point data type
Lower area-ratio bound	Scalar and XY / no-deflation	0.9 / 1.0; the no-deflation momentum increment is projected onto the nonnegative half-line
Positive-area budget	All paths	At most the smaller of 10% of initial whitespace and currently available placeable area per round; negative change is applied before the remaining positive budget is computed
Temporary geometry accounting	Momentum paths	Analytical widths/heights, filler sizes, and pin offsets affect global-placement density and routing feedback only; LEF masters and the original database cell-size array are not modified
Density-proxy clamp	All inflation paths	Enabled in every reported configuration; the proxy dimension in direction q has the common absolute lower bound $\sqrt{2}\delta_q$, where δ_q is the placement-density-bin side length; proxy area is weighted by analytical area divided by proxy area, and XY applies a common post-shaping correction to satisfy both bounds
Geometry-to-DEF transition	All paths	At route-driven GP exit, the optimizer-facing movable sizes and recorded analytical state are rolled back as described in the main text; legalization/DP receives selected center coordinates with the original databases, and DEF lower-left locations are computed with original cell sizes
Directional allocation	XY	$\mathcal{B}_H^t$ and $\mathcal{B}_V^t$ are nonempty in every evaluated map; positive-capacity bins are projected by Π_k^t ; $N_{0.95}$ returns zero for no positive active entry; the indexed weights allocate r_k^t/r_k^{t-1} ; the pre-clamp cumulative scale-factor ratio lies in $[0.8, 1.25]$ and their product is r_k^t ; $H_{k,d}^0 = 0$ and history decay is 0.95
Zero-mass descriptor	Selector	If $\sum C = 0$, $\tau_{\text{hot}} = 0.05$ without evaluating $Q_{0.90}$, and $\rho_{1\%} = \rho_{\text{conn}} = 0$; if both directional means are zero, $\eta_{HV} = 1$
Immediate success condition	Selector / all paths	Zero overflow nets and estimated-short cost at most 10^{-6} ; no analytical geometry update or follow-up GR call is applied
Stagnation window	Momentum paths	Candidate-selection key fails to improve for two rounds and mean absolute area-ratio change is below tolerance
Area-ratio tolerance	Legacy and no-deflation / scalar and XY	0.005 / 0.01; legacy applies the tolerance to its relative area increment
Iteration cap	All finalized operators	Ordinary runs allow termination before a maximum of 15 counted route-optimization rounds; the equal-route control below uses a common $T = 15$ opportunity horizon
Probe accounting	Scalar / no-deflation	Retained scalar probe counts as round 1 / rolled-back probe does not count toward the restarted budget
Candidate-selection key	Scalar	Estimated-short cost, then overflow-net count
Candidate-selection key	XY	Estimated-short cost, then overflow-net count, then $(\bar{R}_H + \bar{R}_V)/2$ as defined in the main text
Candidate-selection key	Legacy and no-deflation	Normalize each wire/via cost by the corresponding first-candidate cost with a 10^{-12} denominator floor; try wire weights (0.93, 0.75, 0.55, 0.30, 0.18, 0) in order; minimize weighted normalized cost, then overflow-net count, then $(\bar{R}_H + \bar{R}_V)/2$; return the first minimizer whose via cost is at most 1.01 times the minimum candidate via cost

The estimated-short cost $S(R) = C_{\text{wire}}(R) + C_{\text{via}}(R)$ is computed from the global router's capacity, wire-demand, and via-demand maps. Its terms are the layer-scaled positive wire-overflow area and the via demand at routing locations where wire demand exceeds capacity, respectively.

Input-transformation audit

The processed inputs are used only for matched comparisons within each suite. The checks below establish transformation coverage and exact correspondence between the documented scripts and generated files; they do not establish equivalence to the original constrained benchmark problems or official contest scores.

Table S2: Input transformations applied to the compatibility-processed ISPD 2015 suite.

Transformation	Verified coverage	Validity interpretation
LEF property/header normalization; unsupported TOPOFSTACKONLY removal	20/20; 15/20	Parser normalization; no claim of representation equivalence
DEF VIAS migration into generated LEF	20/20	Definitions moved rather than discarded
DEF NONDEFAULTRULES migration into generated LEF	20/20	Definitions moved rather than discarded
Multi-layer TRACKS splitting	20/20	Global-router parser compatibility
Routed special-net segment rewriting	20/20	Flow compatibility; an open special net may remain
EXCEPTPGNET added to applicable LEF obstructions	14/20	Changes selected power/ground checks
EXCEPTPGNET added to applicable DEF routing blockages	5/20	Changes selected power/ground checks
REGIONS/GROUPS removal	9/20	Removes fence constraints; affected designs carry † in the main manuscript’s primary-results table
Fixed-component coordinate snapping	5/20	Alters specified physical coordinates
LEF pin-rectangle grid snapping	4/20	Alters specified pin geometry
SOURCE DIST removal	5/20	Superblue parser compatibility
DEFAULT insertion for VIA declarations	5/20	Superblue routing compatibility
placement.constraints copy	20/20 byte-identical	Sidecar placement-constraint file preserved

The nine fence-removal designs are `des_perf.a`, `des_perf.b`, `edit_dist.a`, `matrix_mult.b`, `matrix_mult.c`, `pci_bridge32.a`, `pci_bridge32.b`, `superblue11.a`, and `superblue16.a`. They carry a dagger in the main manuscript’s primary-results table. All variants receive the same processed input for a given design. Constraint removal, special-net rewriting, and geometry correction can change the feasible placement or checked routing interactions, so the audit supports within-suite matched development-family comparison but not held-out validation or sign-off completeness.

Table S3: Exact reconstruction audit for the eight compatibility-processed ICCAD 2017 instances.

Design	Fence sections removed	Snapped fixed/COVER locations	DEF exact	LEF exact
<code>des_perf.b.md1</code>	yes	0	yes	yes
<code>des_perf.b.md2</code>	yes	0	yes	yes
<code>edit_dist.1.md1</code>	no	0	yes	yes
<code>edit_dist.a.md2</code>	yes	6	yes	yes
<code>fft.2.md2</code>	no	0	yes	yes
<code>fft.a.md2</code>	no	0	yes	yes
<code>fft.a.md3</code>	no	0	yes	yes
<code>pci_bridge32.a.md1</code>	yes	0	yes	yes

The ICCAD audit compares each archived original design with the generated single-LEF/single-DEF input. Every generated DEF exactly matches removal of complete REGIONS/GROUPS sections followed by manufacturing-grid correction; every generated LEF exactly matches the scripted technology/cell-library merge. All eight designs use a 5-DBU manufacturing grid. These exact checks establish implementation-to-output correspondence, not equivalence to the original multi-deck legalization task.

Equal-route-opportunity control provenance

Table S4: Publicly releasable provenance for the equal-route-opportunity control.

Artifact	Public identifier, SHA-256, or setting
Experiment commit	c7c43fdaf309543a07bb72050b79bdd709e79324
Protocol	b8996d84a8d330d80a5351d56cd7ac2d38c51f0a8c813cfa2d280c23b431c238
E1-EQ CSV	dc0554e635a2e9e291915f316429727ecfc86aad67b50dee15ca26b27445c393
E4-EQ CSV	5a63f22dc71ffd6c62b8f19230c33cd42e130f78c19e8fd782ea397a40aed3ef
Public budget audit CSV	177b3028d2c6ce3750bd51440b528074fcea0d4f73527e06403f48d0b266a355
Public verification report	133ccf1355ca5364e83fc34512e498636a30a63bfff9423f721b67b62273f57a
Innovus release	v22.30-s003_1
Execution controls	seed 0; deterministic DEF generation

A counted opportunity round consists of one placement-refinement stage followed by one GR evaluation; **Route Iter** denotes the corresponding logged event. The initial trigger route precedes these rounds and is not counted, but it is retained as an evaluated GR candidate. Both equal-route variants use $T = 15$ counted opportunity rounds. If ordinary operator termination is reached before T , cell-size actuation is disabled and the current analytical size state is frozen, while refinement and its GR evaluation continue for the remaining opportunity rounds. The GR evaluation associated with round 15 is the final candidate-selection evaluation. Including the initial trigger route, each variant–design pair therefore has 15 **Route Iter** events and 16 GR evaluations and candidates. Selector decisions, pre-freeze actuation, ordinary termination conditions, and candidate-selection keys are unchanged.

Table S5: Opportunity-count audit. Each row represents two passing variant–design pairs.

Design	E1-EQ		E4-EQ		Audit					
	Route	Iter	GR	calls	Route	Iter	GR	calls	E1-EQ	E4-EQ
des_perf_b.md1		15		16		15		16	PASS	PASS
des_perf_b.md2		15		16		15		16	PASS	PASS
edit_dist_1.md1		15		16		15		16	PASS	PASS
edit_dist_a.md2		15		16		15		16	PASS	PASS
fft_2.md2		15		16		15		16	PASS	PASS
fft_a.md2		15		16		15		16	PASS	PASS
fft_a.md3		15		16		15		16	PASS	PASS
pci_bridge32_a.md1		15		16		15		16	PASS	PASS

Table S6: Per-design secondary outcomes in the equal-route-opportunity control. Time is the per-DEF arithmetic mean of three detailed-routing runs. Ratio is E4-EQ/E1-EQ.

Design	DRWL			DR vias			DR time/s		
	E1-EQ	E4-EQ	Ratio	E1-EQ	E4-EQ	Ratio	E1-EQ	E4-EQ	Ratio
des_perf_b.md1	1424955.165	1441258.980	1.011442	561600	565707	1.007313	788	926	1.175127
des_perf_b.md2	1513532.050	1541317.390	1.018358	569251	570038	1.001383	706	731	1.035411
edit_dist_1.md1	4596803.815	4410966.935	0.959573	856917	875666	1.021880	1770	1882	1.063277
edit_dist_a.md2	5729955.555	5134226.820	0.896033	1019745	946151	0.927831	1408	1339	0.950994
fft_2.md2	624352.435	628922.525	1.007320	198407	204403	1.030221	325	1768	5.440000
fft_a.md2	1039902.425	982295.115	0.944603	190851	195171	1.022635	709	833	1.174894
fft_a.md3	957374.450	972266.085	1.015555	191675	194047	1.012375	883	803	0.909400
pci_bridge32_a.md1	366494.680	371684.570	1.014161	141346	144064	1.019229	340	370	1.088235

Table S7: Placement time in the equal-route-opportunity control, matched by design to the corresponding placement-run summary. Ratio is E4-EQ/E1-EQ.

Design	E1-EQ/s	E4-EQ/s	Ratio
des_perf_b.md1	36.093	38.730	1.073068
des_perf_b.md2	35.870	38.007	1.059566
edit_dist_1.md1	45.233	44.215	0.977500
edit_dist_a.md2	47.617	45.542	0.956415
fft_2.md2	19.624	20.854	1.062669
fft_a.md2	22.295	24.087	1.080368
fft_a.md3	21.662	24.267	1.120259
pci_bridge32_a.md1	20.499	22.346	1.090088

For the DRC aggregate, the wins-only geometric change is -28.73% and the losses-only change is $+14.40\%$. Removing `edit_dist_1.md1` changes the seven-design geometric aggregate to $+0.04\%$, while removing `fft_2.md2` changes it to $+0.81\%$. These influence calculations are secondary sensitivity analyses rather than alternate primary endpoints.

Each DEF was routed independently three times using the same Innovus configuration as the ISPD 2015 flow. DRC, DRWL, and via count were invariant across the three runs, and detailed-routing time is reported as their arithmetic mean. Raw Innovus logs are not publicly available, so repeat-level time dispersion cannot be assessed. Equal route-optimization and global-routing counts match routing opportunity but do not imply equal compute, wall time, work per call, or state trajectory. The accompanying public repository provides an artifact index, machine-readable final tables, public transformation and opportunity-count audits, and a checksum manifest; it excludes restricted inputs, generated and routed DEFs, per-repeat reports, and raw Innovus logs.